



Deep-Dive into OpenStack Identity & Dashboard Services

Architectural Fundamentals, RBAC Security, and Horizon Operations

KEYSTONE · HORIZON · RBAC

Session Agenda

A structured journey from identity fundamentals through security architecture to hands-on operational practice.

01

Foundations

Keystone Authentication vs. Authorization – core entities, scopes, and identity concepts

03

Visualization

Horizon architecture, multi-tenant views, and the dashboard operations workflow

02

Security

Role-Based Access Control (RBAC), system vs. project scoping, and token workflows

04

Hands-on Labs

Lab 1: Keystone CLI domain & identity management – Lab 2: Horizon GUI tenant provisioning & VM launch

Keystone: Authentication vs. Authorization

Authentication (AuthN)

"Who are you?" Keystone verifies identity via credentials, X.509 certificates, or federated external identity providers (IdP). Authentication operates globally across the entire cloud.

Authorization (AuthZ)

"What can you do?" Once identity is confirmed, Keystone determines the user's permissions strictly within a specific project or domain scope – never globally.

Core Entities

- **Users** hold credentials and identity
- **Projects** hold and isolate cloud resources
- **Roles** bridge users to projects, granting specific permitted actions



Key isolation principle: AuthN is cloud-wide; AuthZ is always bounded by project or domain scope.



Role-Based Access Control (RBAC) & Next-Gen Policies

Modern OpenStack moves beyond coarse-grained roles with a formal scoping model that separates operator-level infrastructure control from tenant-level resource access.

Classic RBAC

Traditional model relied on three broad roles: **admin**, **member**, and **reader** – powerful but lacking precision for multi-tenant clouds.

System Scope

Reserved for cloud infrastructure operators. Grants access to global resources – managing hypervisor nodes, hardware inventory, and cross-tenant configurations.

Project Scope

Bounded within tenant limits. Application developers use this scope to spin up VMs, configure networks, and manage storage – without visibility into other tenants.

The Reader Role

A key compliance enabler: grants full **read-only visibility** into configurations without any risk of data alteration – ideal for auditors and security reviewers.

The Service Catalog: The Cloud Routing Matrix

Keystone serves as the authoritative service registry for every OpenStack API component. Services never hardcode IP addresses – they query Keystone dynamically to resolve current API endpoints at runtime.

Public Endpoint

Accessible by external end-users, tenant applications, and internet-facing clients. This is the URL published to operators and consumers of the cloud.

Internal Endpoint

Used for secure, high-speed controller-to-controller communication between OpenStack services (Nova, Neutron, Cinder) across the management network.

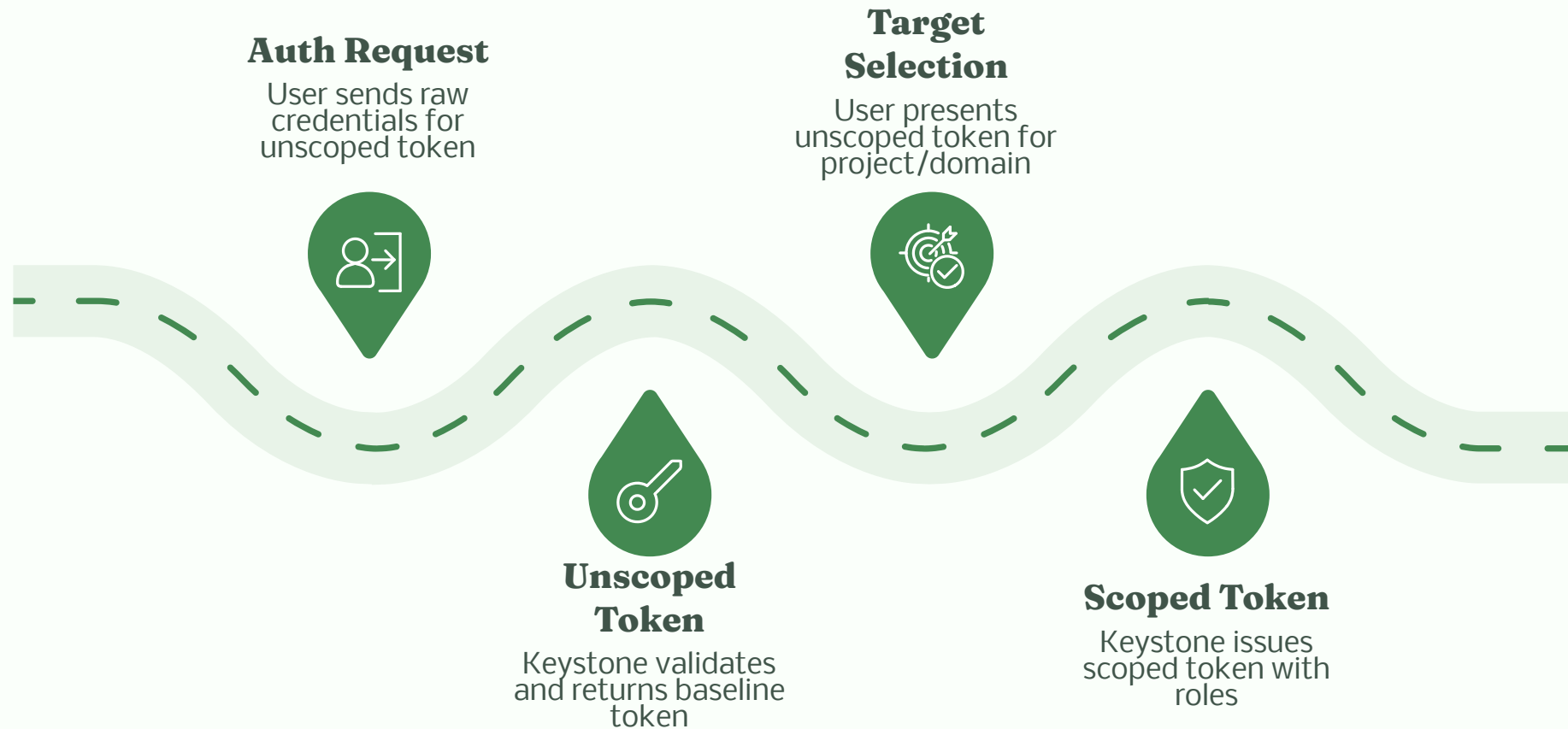
Admin Endpoint

A restricted network path reserved exclusively for privileged management operations. Typically bound to the admin network and not exposed externally.

 Every registered service – Nova, Neutron, Cinder, Glance, and more – maintains all three endpoint tiers simultaneously in the catalog.

The Anatomy of a Token Workflow

Every API interaction in OpenStack is governed by a structured token lifecycle. Understanding this flow is essential for debugging authentication failures and designing secure automation pipelines.



The scoped token is the operational passport – it encodes the user's identity, project context, and assigned roles in a single signed artifact. Nova and Neutron validate every inbound token against Keystone before executing any resource operation.

Horizon Architecture & Integration Layer

Stateless Django Application

Horizon stores **no local state or database**. It is a pure Django web application – every user action is a real-time translation of UI interactions into OpenStack REST API calls. This makes it inherently scalable and easy to replace.

Extensible Plugin System

Dynamic dashboard panels can be added without modifying core code. Octavia load balancer panels, Designate DNS management, and custom operator panels all integrate seamlessly via the plugin architecture.

High Availability Deployment

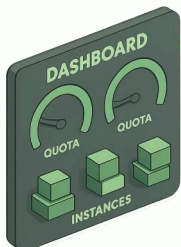
Horizon sits behind **Apache or Nginx** web servers and scales horizontally across multiple controller nodes. A Virtual IP (VIP) managed by HAProxy or Keepalived ensures continuous availability across node failures.

Direct API Broker

Every mouse click or form submission instantly maps to a corresponding OpenStack REST API call – no middleware buffering or delayed synchronization. What you see reflects live infrastructure state.

Multi-Tenant Visual Isolation: Project vs. Admin Views

Horizon enforces role-based UI segmentation – the view a user sees is determined entirely by their assigned Keystone roles and project membership.



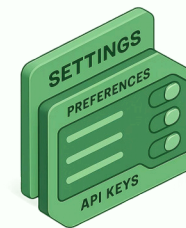
Project View

Tailored for self-service tenant consumers. Displays **abstract quotas**, running compute instances, private subnet topology, and attached block storage volumes – scoped exclusively to the user's own project.



Admin View

Visible only to users with global administrative roles. Exposes **physical hypervisor health**, compute node memory overcommit ratios, global floating IP allocation pools, and aggregate resource usage across all tenants.

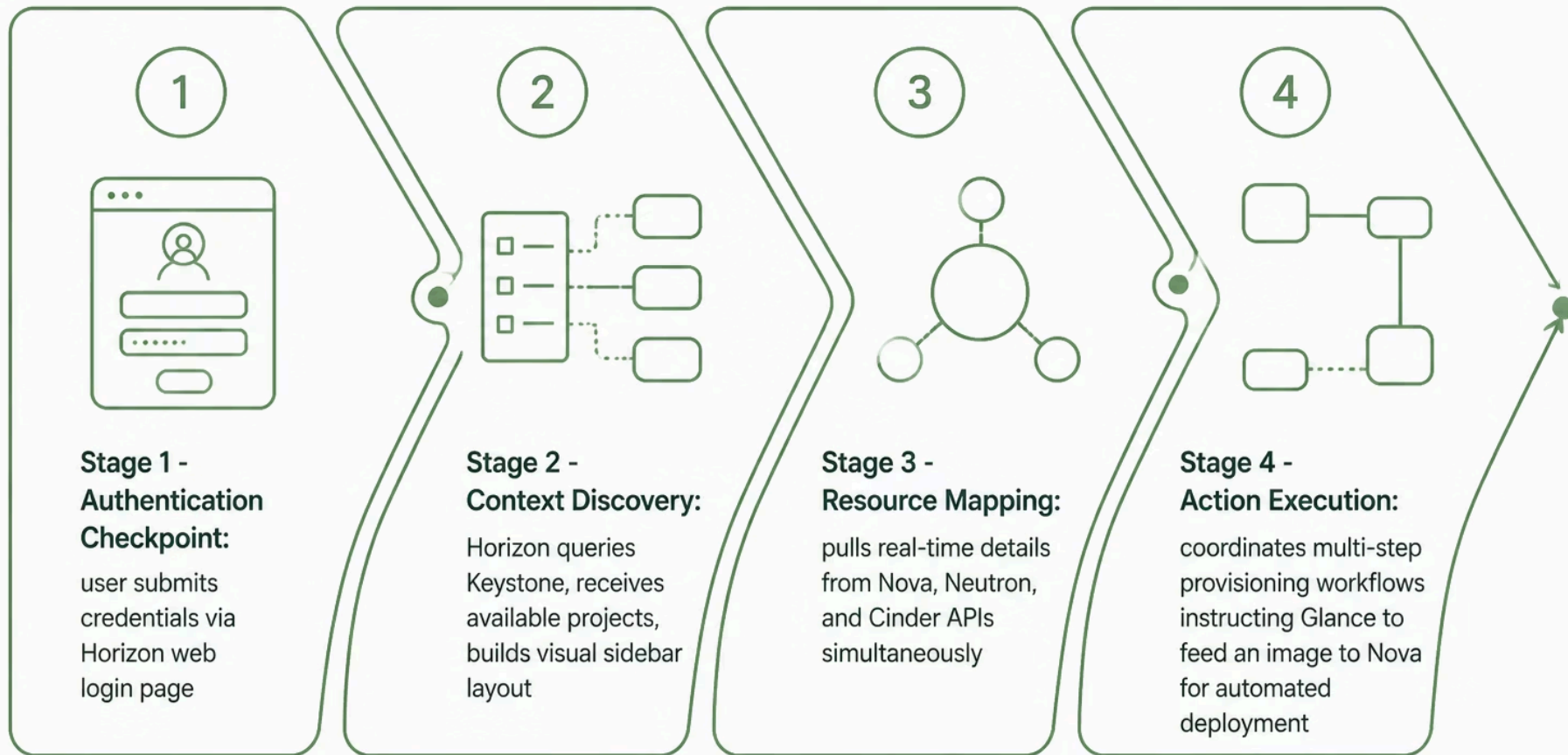


Settings Panel

Contextual user management for **language preferences**, time zone configuration, and secure API credential (RC file) downloads – enabling CLI and SDK access alongside the GUI.

Unified Dashboard Operations Workflow

Horizon orchestrates a coordinated sequence of API calls across multiple OpenStack services to deliver a seamless self-service experience from login to full VM provisioning.



This workflow highlights Horizon's role as a **pure orchestration layer** – it holds no state of its own, but coordinates live data from four or more OpenStack services in every page render cycle.

Hands-On Labs

Reinforce architectural concepts with direct CLI and GUI practice across both identity and dashboard services.

1

Lab 1: Keystone CLI — Domain & Identity Management

Create domains, projects, users, and role assignments using the `openstack` CLI. Validate scoped token issuance and verify policy enforcement with both **member** and **reader** roles. Practice generating and inspecting token payloads.

- Create a new domain and project hierarchy
- Assign system-scoped vs. project-scoped roles
- Request, scope, and introspect tokens via CLI

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Lab 2: Horizon GUI — Tenant Provisioning & VM Launch

Log into Horizon as both a project member and an admin. Explore the visual differences in available panels and resource visibility. Launch a VM instance end-to-end using the guided provisioning wizard.

- Compare Project View vs. Admin View side-by-side
- Provision a VM: image → flavor → network → security group
- Download the OpenStack RC file and verify CLI equivalence